

Five API Models: Final Clean Comparison

1 Data Source

This report analyzes the final clean result set in `results/api_models_final_clean`. It contains 44 benchmarks for each of five API models: `gpt-4o`, `gpt-4.1`, `gpt-5`, `claude-sonnet-4-5`, and `gemini-3.5-flash`. The result set contains 220 benchmark files and 4,345 sample results in total, with no sample-level errors.

The figures and CSV tables were generated with:

```
python comparison\build_report.py -results-dir results\api_models_final_clean
-output-dir comparison\output\five_api_models_final_clean -bootstrap-samples
5000
```

2 Overall Summary

Table 1: Overall model summary across 44 benchmarks.

Model	Mean score	Median score	Mean normalized	Mean rank
gemini-3.5-flash	0.585	0.725	0.790	1.82
gpt-4.1	0.559	0.625	0.668	2.18
gpt-5	0.519	0.625	0.567	2.66
gpt-4o	0.509	0.550	0.573	2.70
claude-sonnet-4-5	0.419	0.393	0.254	3.82

Table 2: Family leaders by mean score.

Family	Leader	Mean score
Captioning	gpt-4o	0.170
Question answering	gemini-3.5-flash	0.933
Labeling	gemini-3.5-flash / gpt-4.1	0.738
Detection	gemini-3.5-flash	0.186
Image modification VQA	gemini-3.5-flash	0.925
Image preference	gpt-5	0.750
Aesthetic rating	gemini-3.5-flash	0.817
Prompt reconstruction	claude-sonnet-4-5	0.967

3 Score Figures

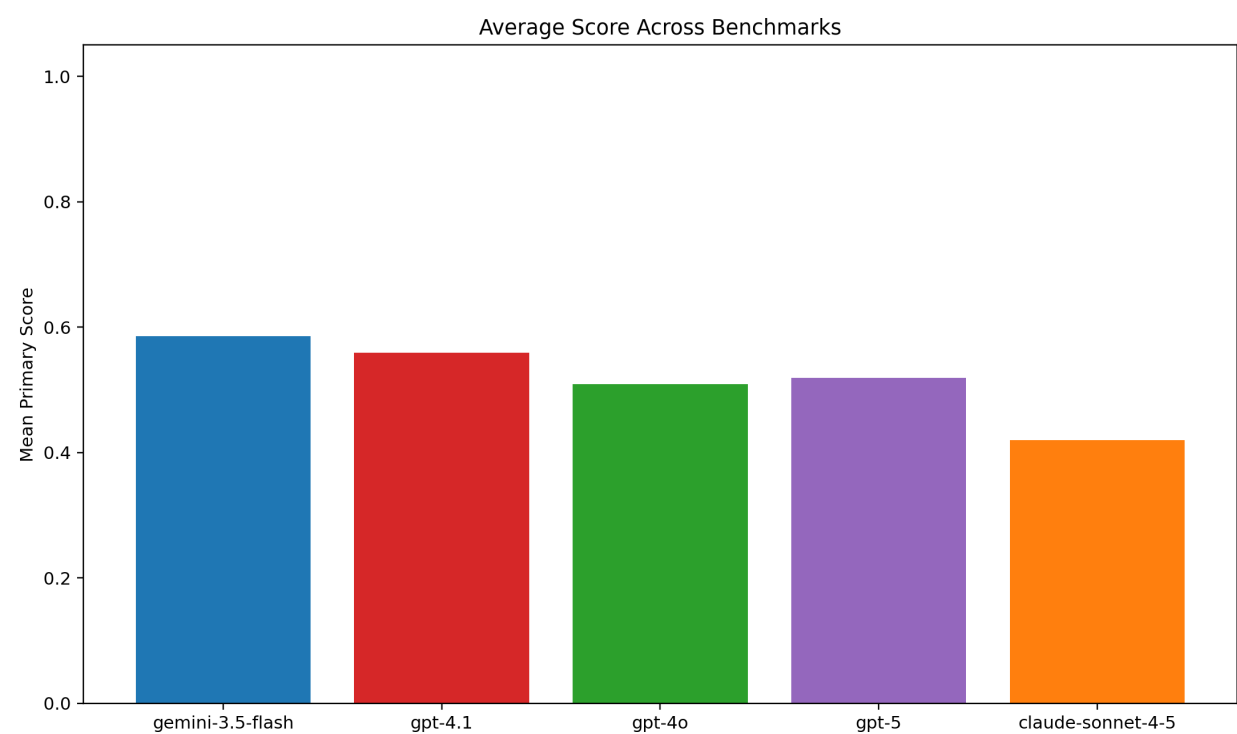


Figure 1: Mean primary score across all matched benchmarks.

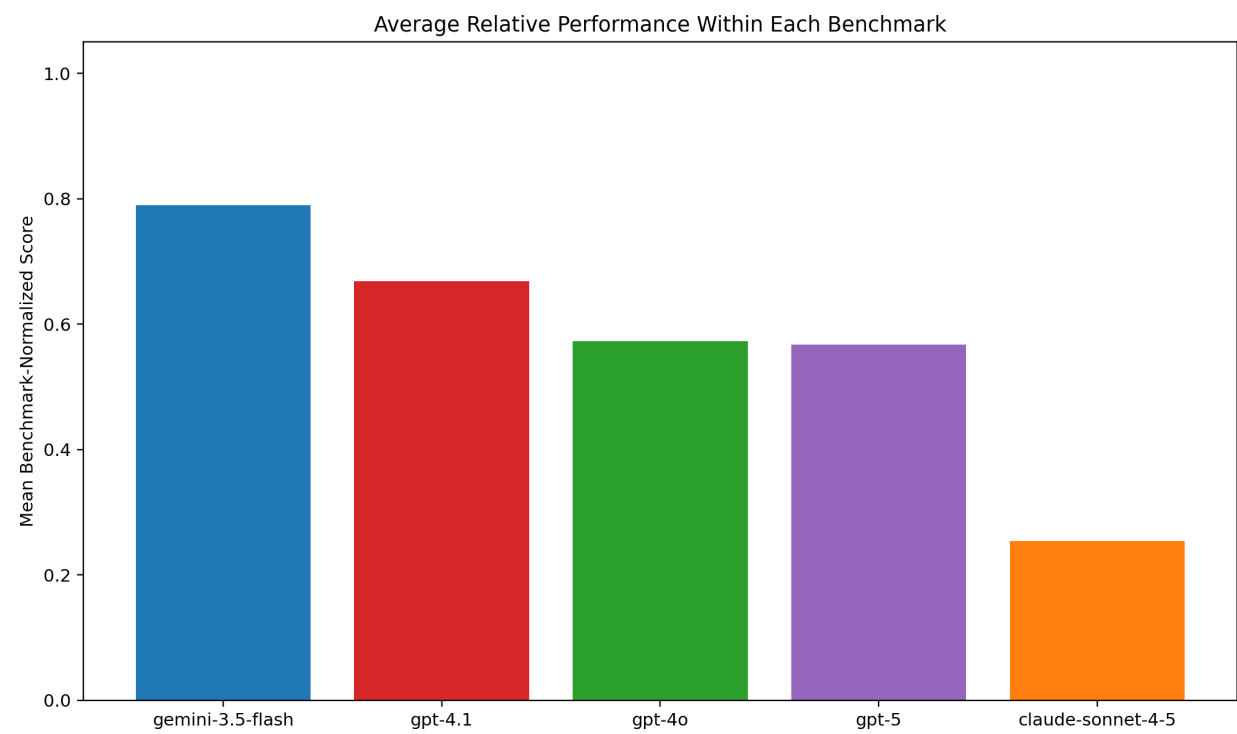


Figure 2: Mean score after normalizing each benchmark between the best and worst model.

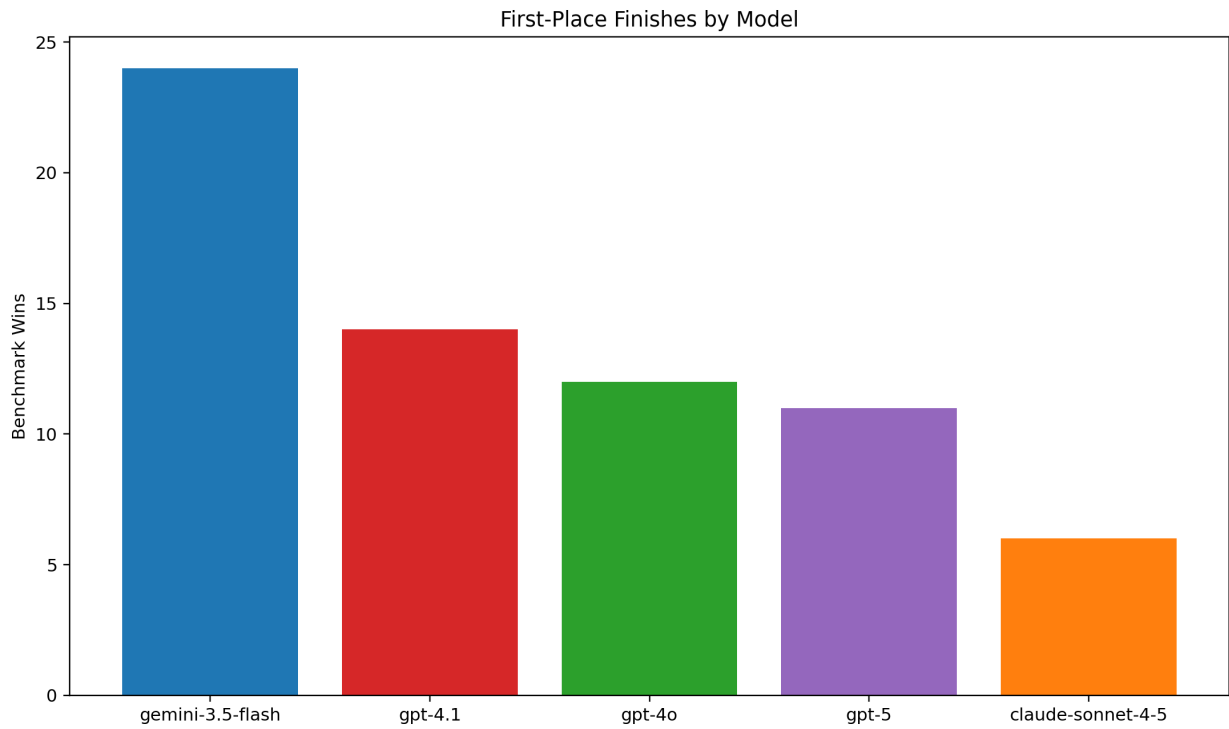


Figure 3: Number of benchmarks where each model finished in first place.

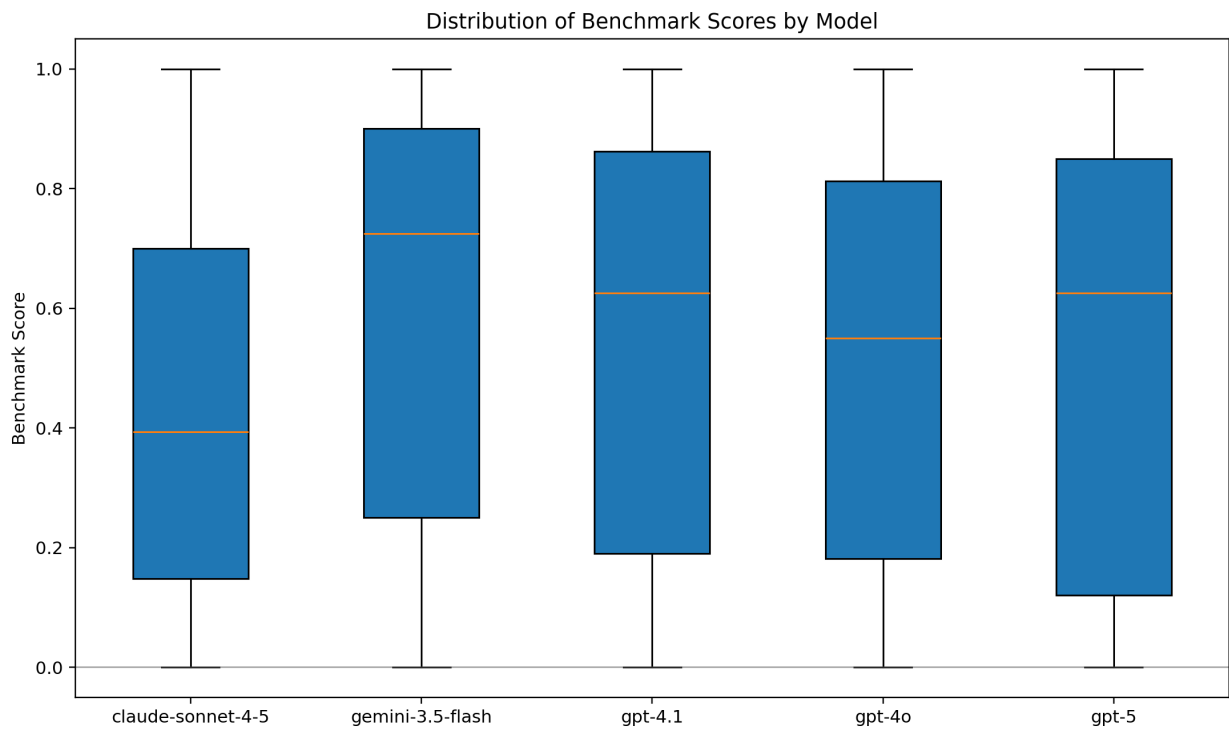


Figure 4: Distribution of primary benchmark scores for each model.

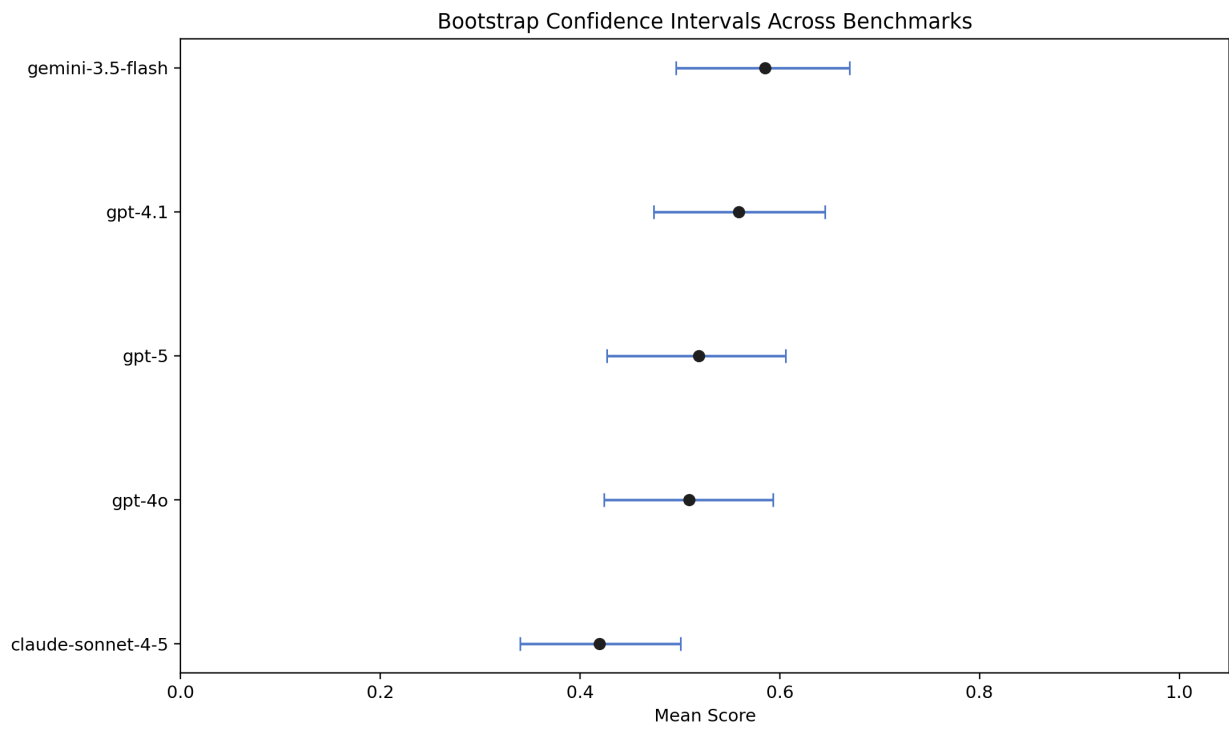


Figure 5: Bootstrap confidence intervals for each model’s mean benchmark score.

4 Benchmark-Level Figures

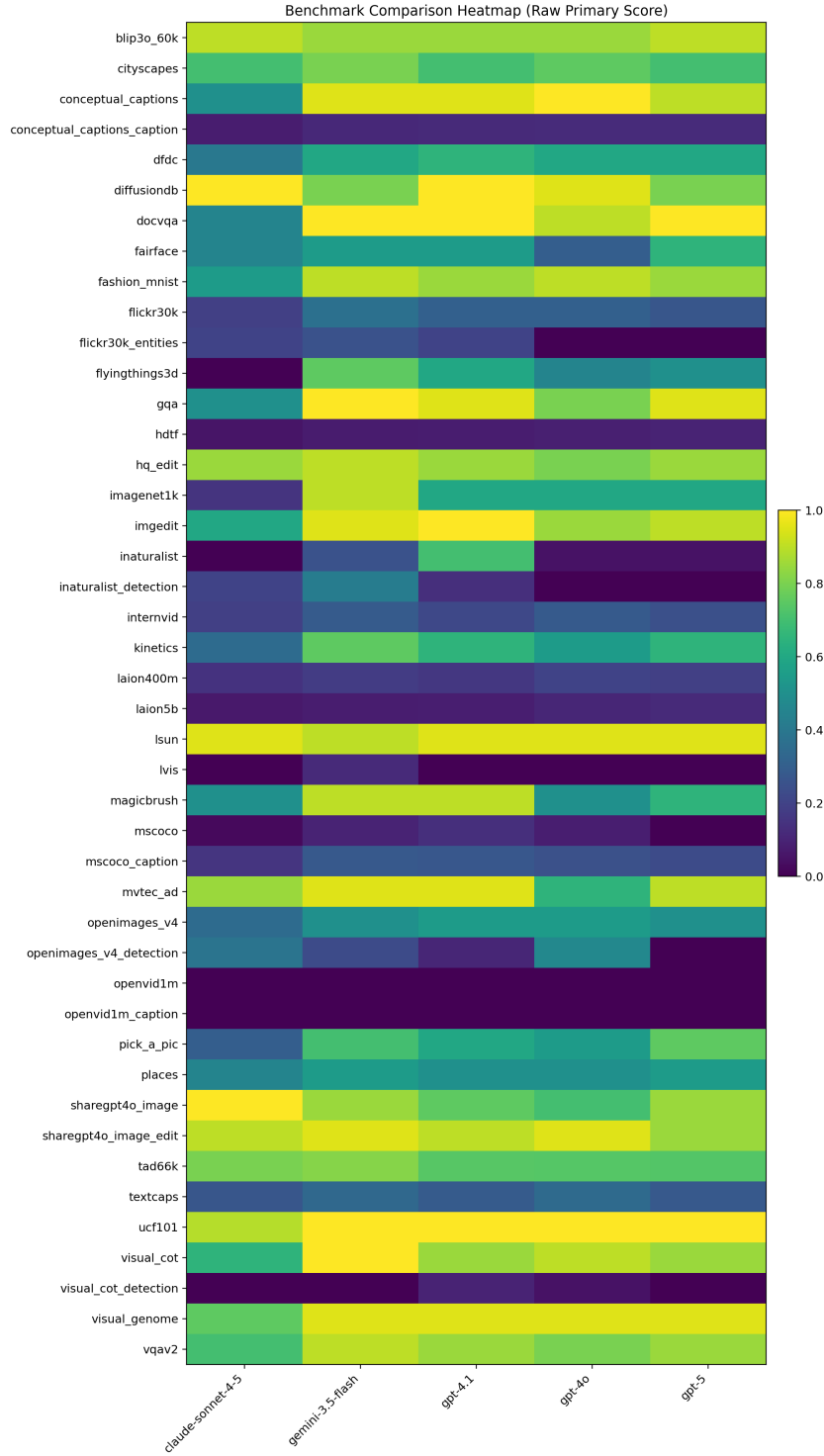


Figure 6: Raw primary score heatmap by benchmark and model.

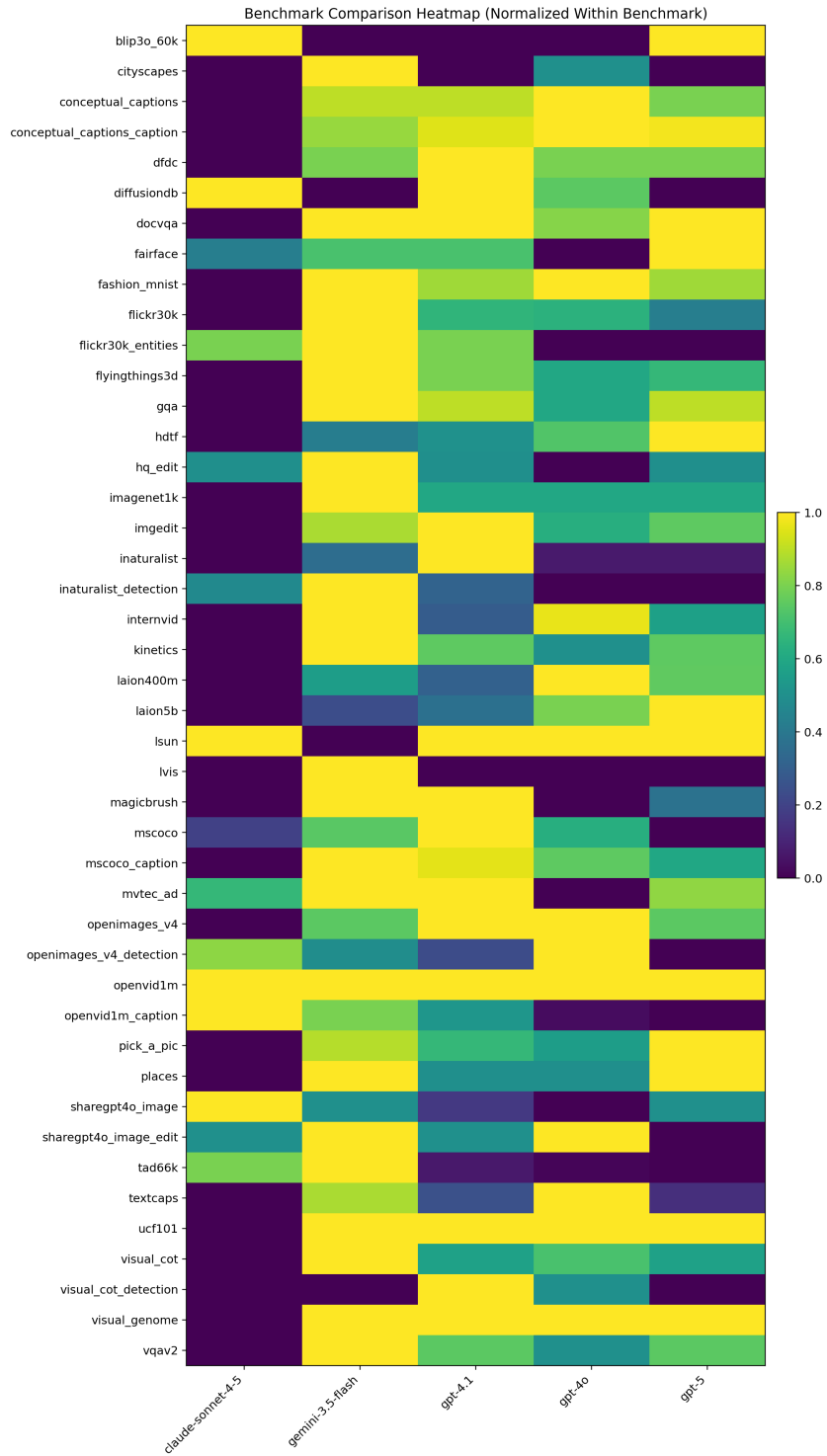


Figure 7: Benchmark-normalized score heatmap by benchmark and model.

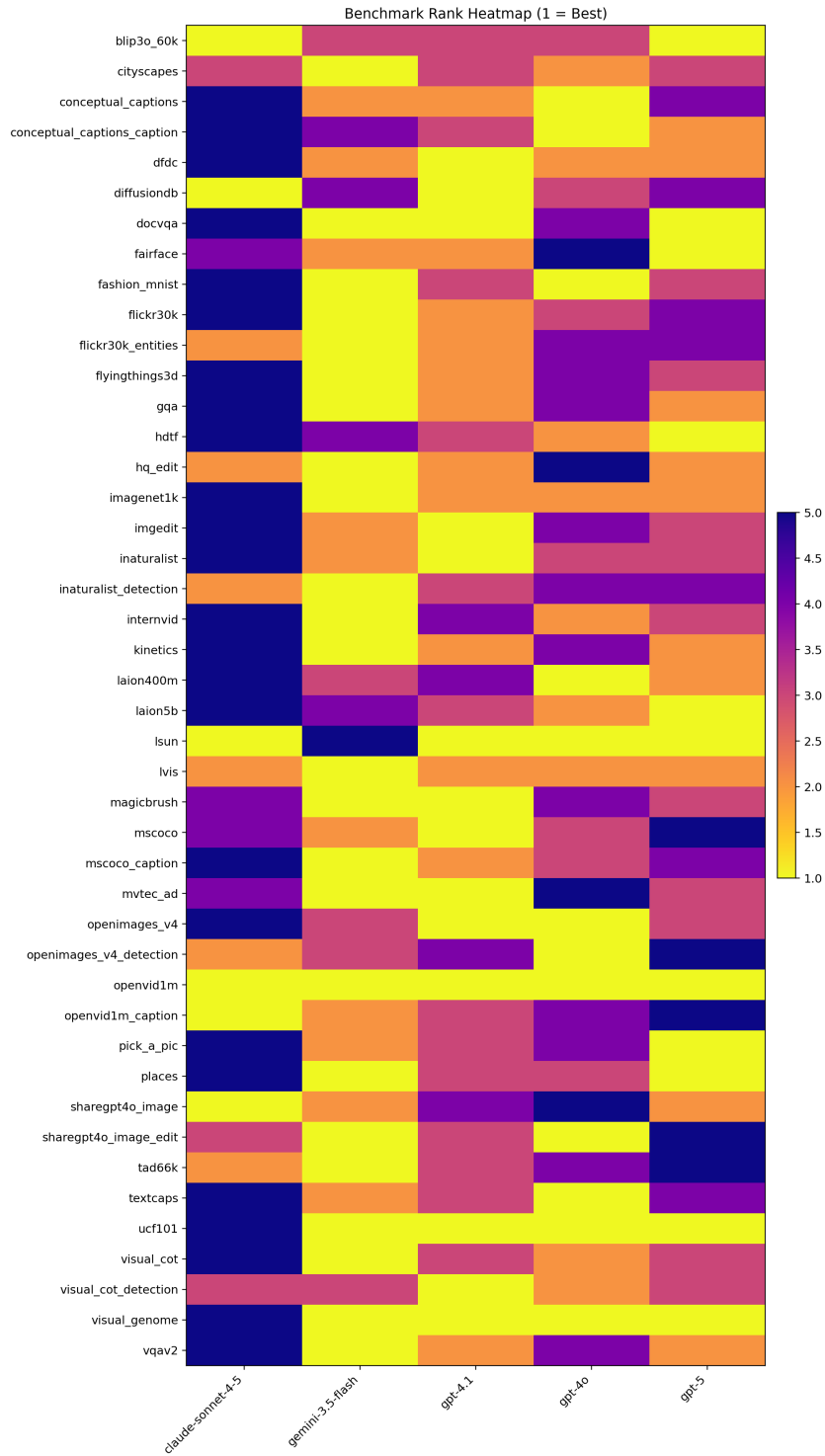


Figure 8: Rank heatmap by benchmark, where 1 is best.

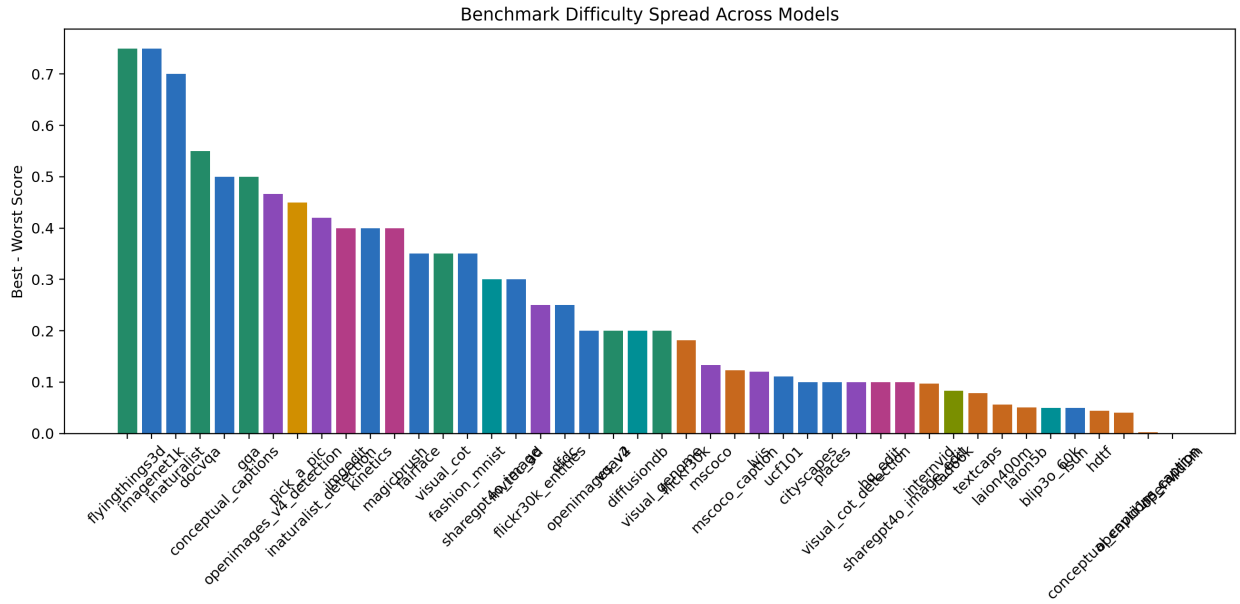


Figure 9: Difference between the best and worst model score for each benchmark.

5 Family Figures

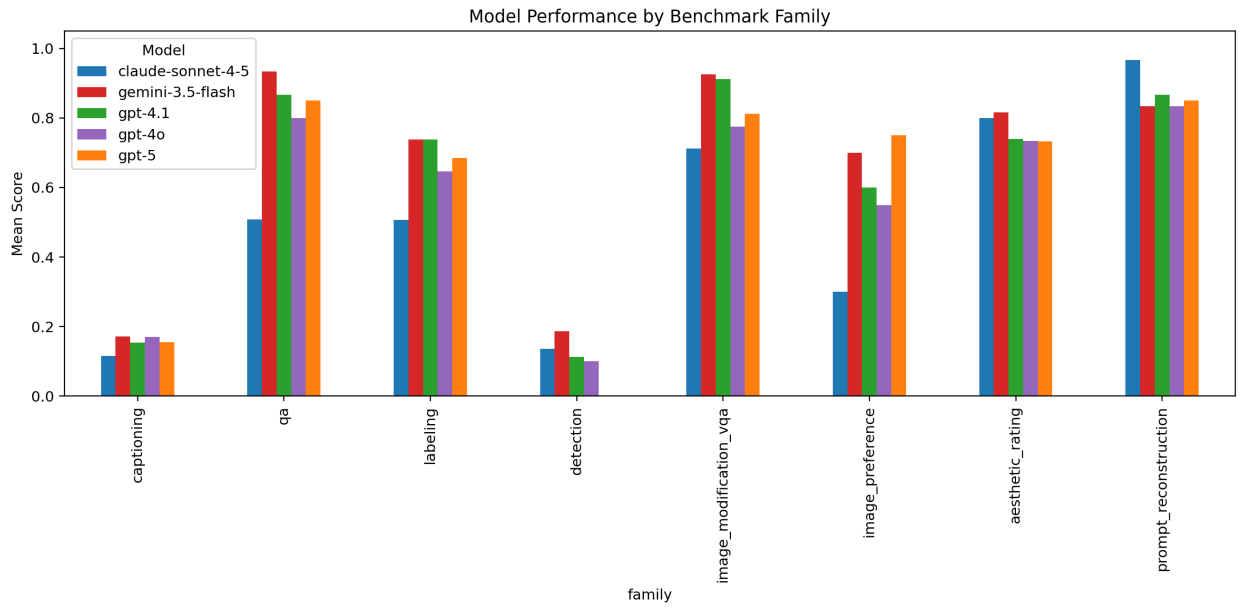


Figure 10: Mean model score grouped by benchmark family.

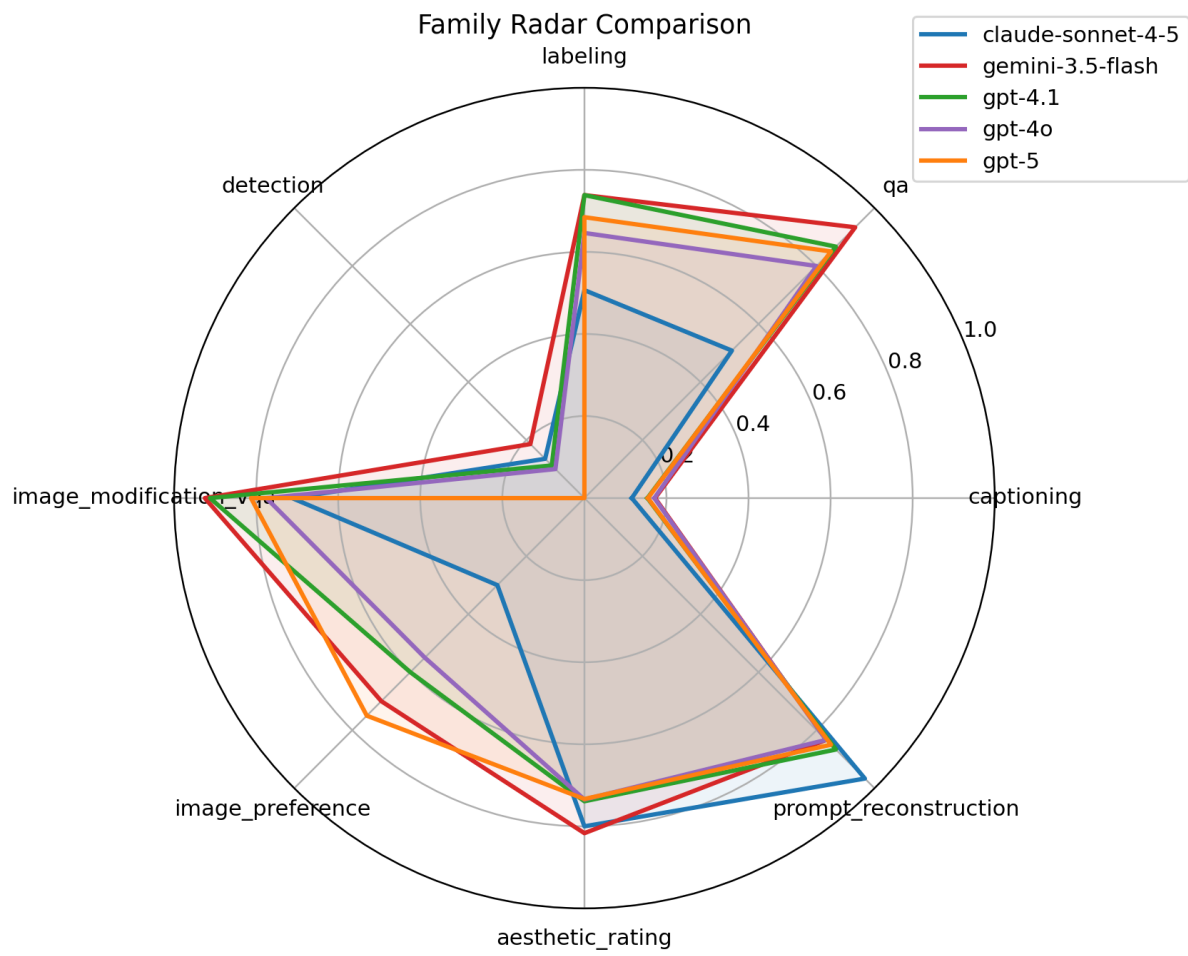


Figure 11: Radar plot of model performance across benchmark families.

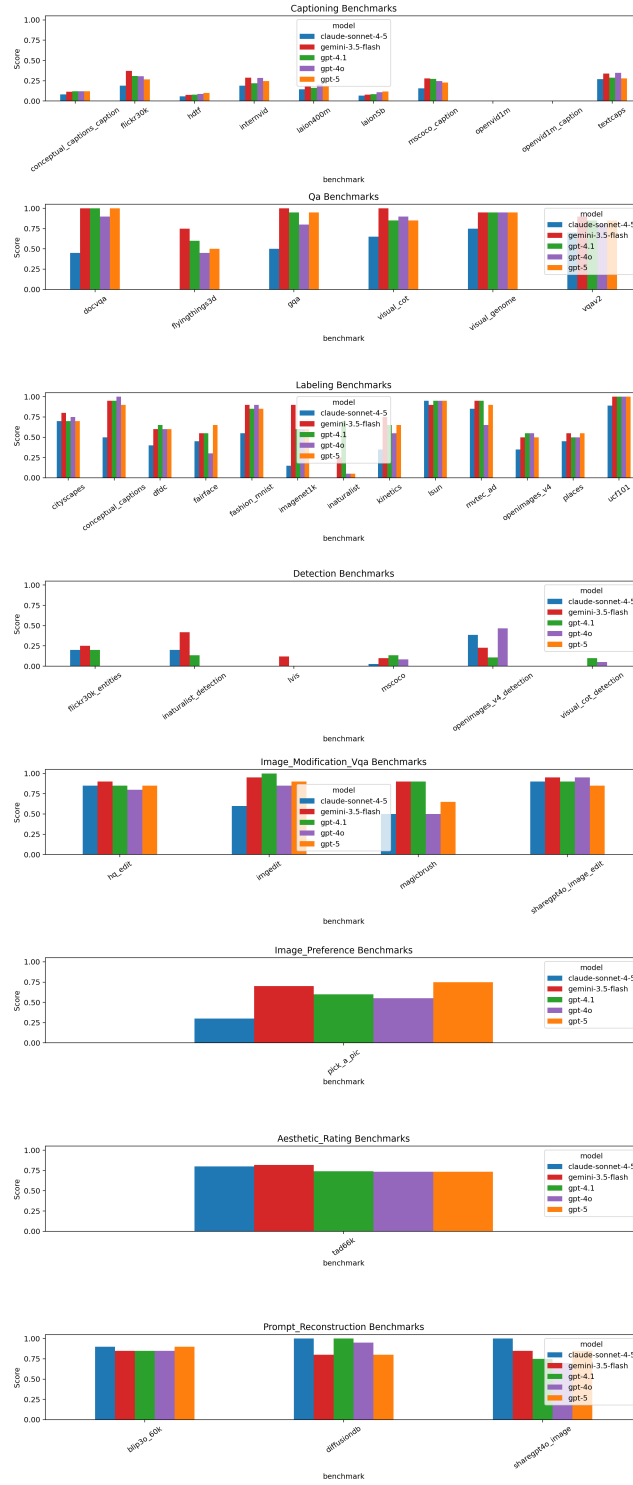


Figure 12: Per-family benchmark score bars.

6 Pairwise Figures

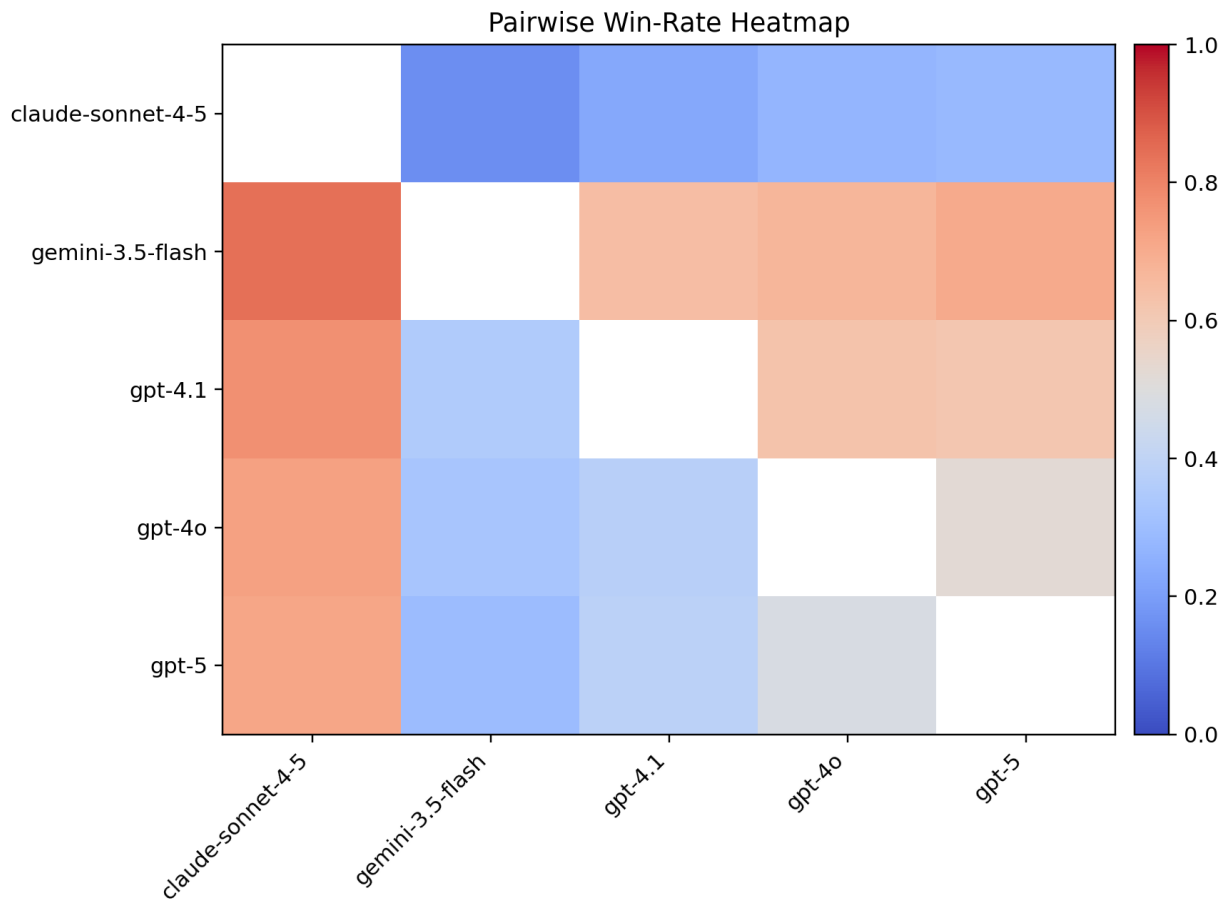


Figure 13: Pairwise model win-rate heatmap across benchmarks.

7 Telemetry Figures

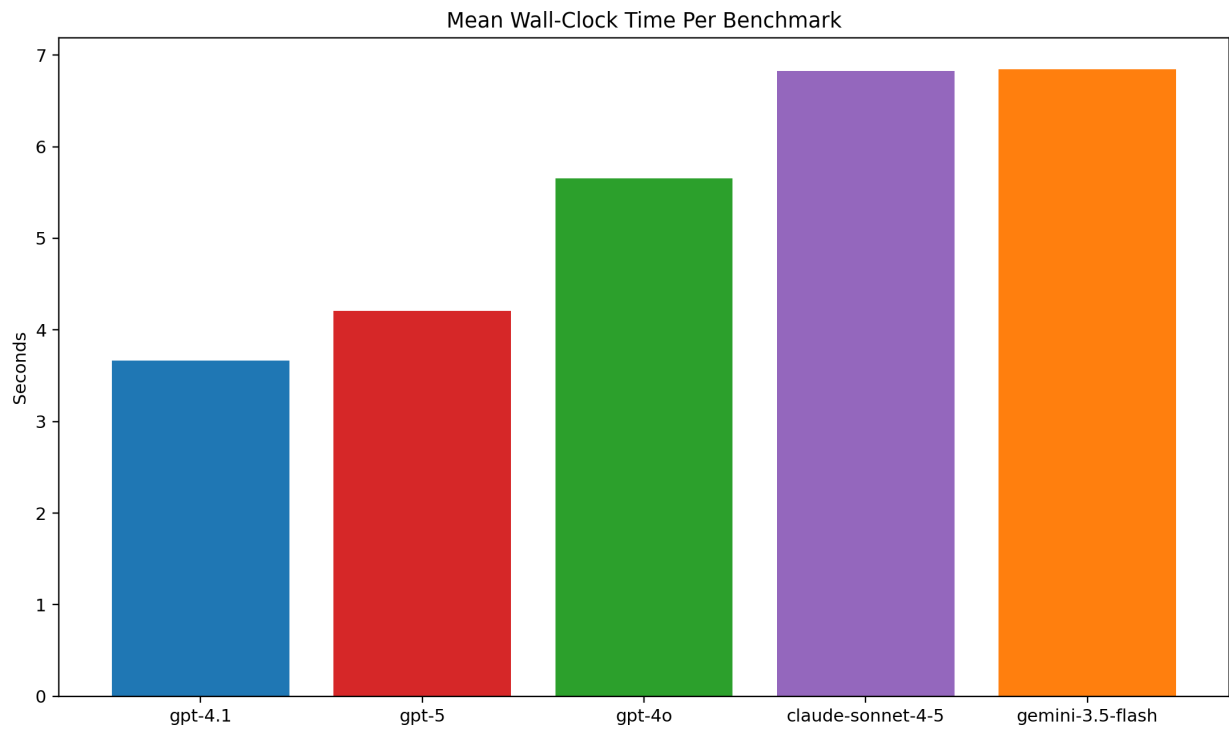


Figure 14: Mean wall-clock time per benchmark by model.

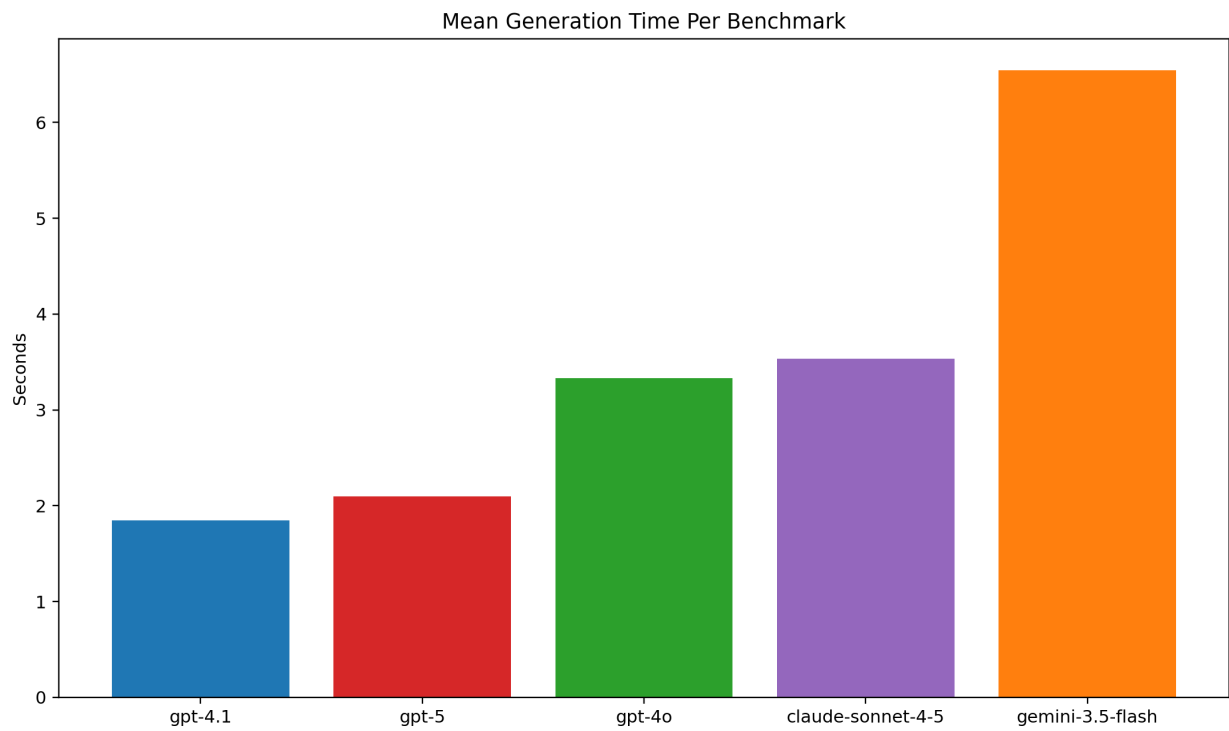


Figure 15: Mean generation time per benchmark by model.

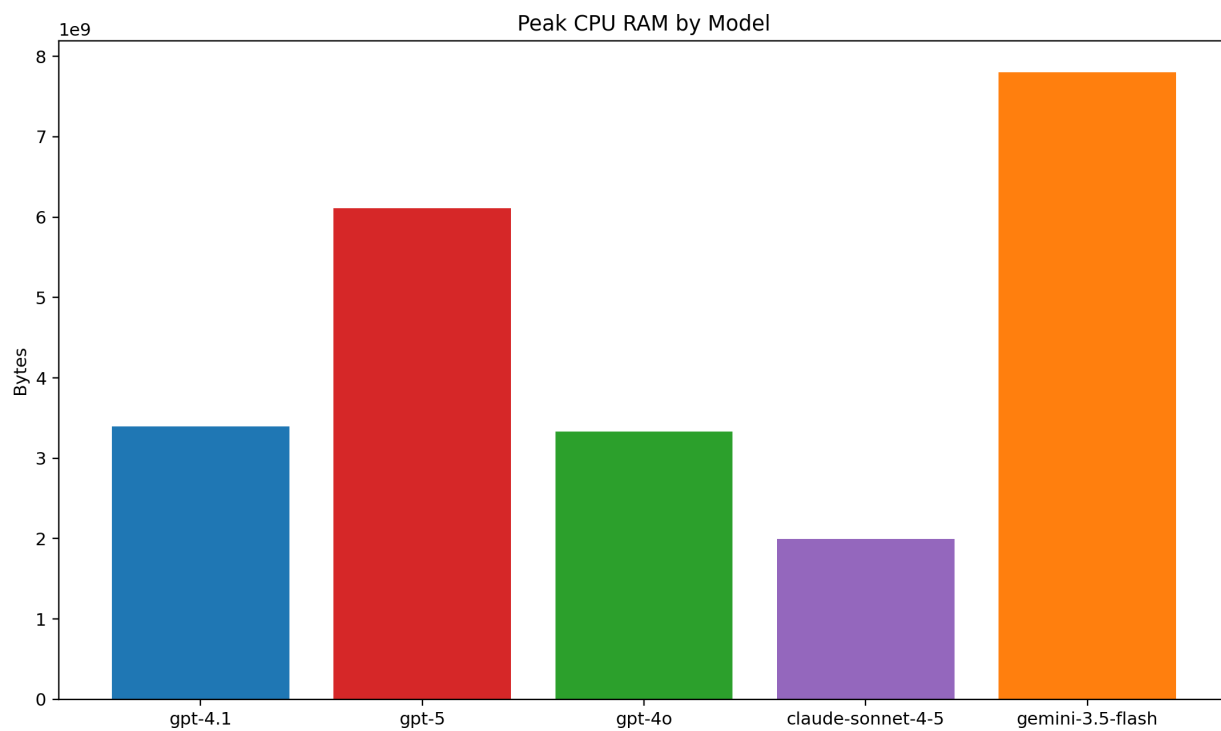
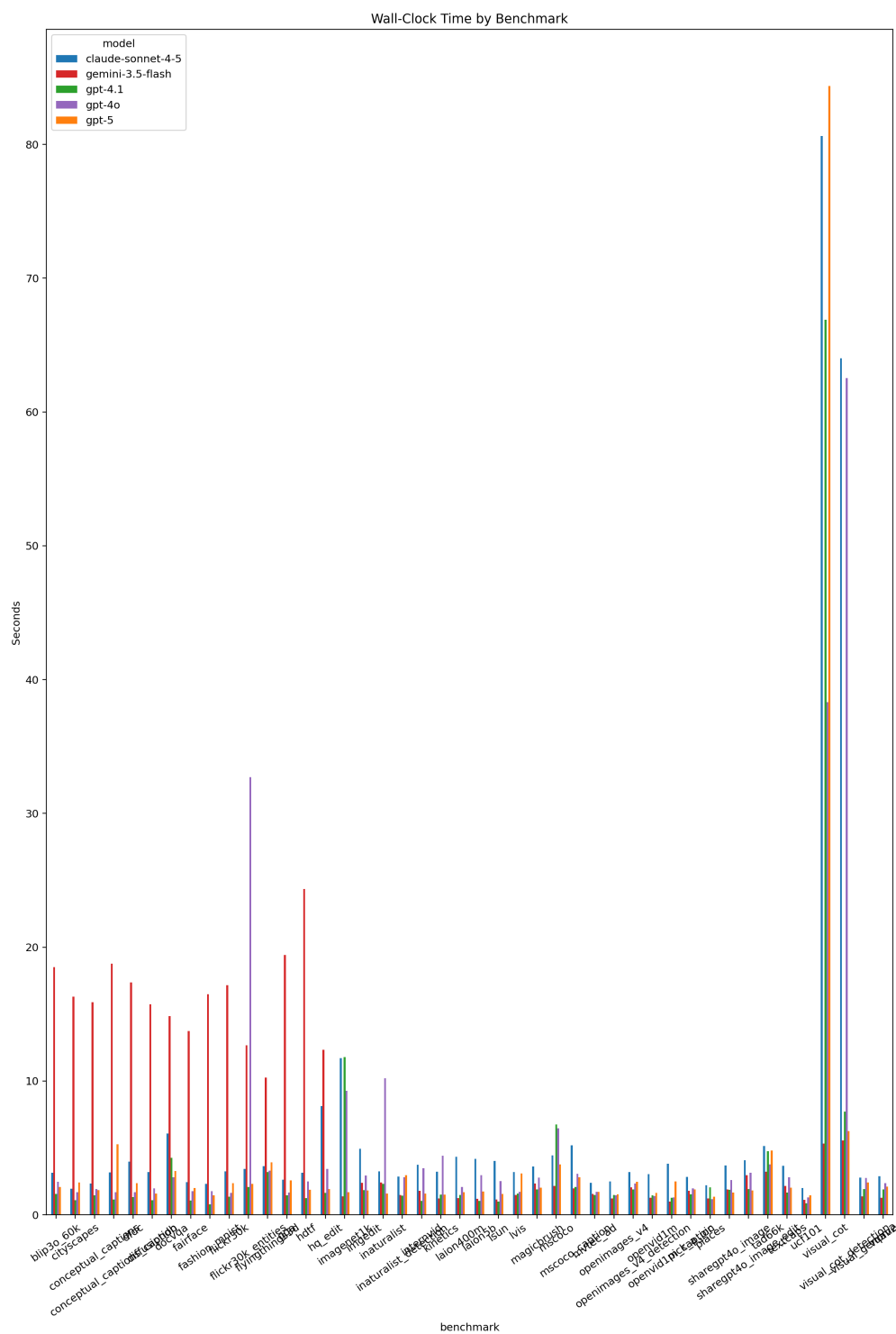


Figure 16: Peak CPU RAM by model.



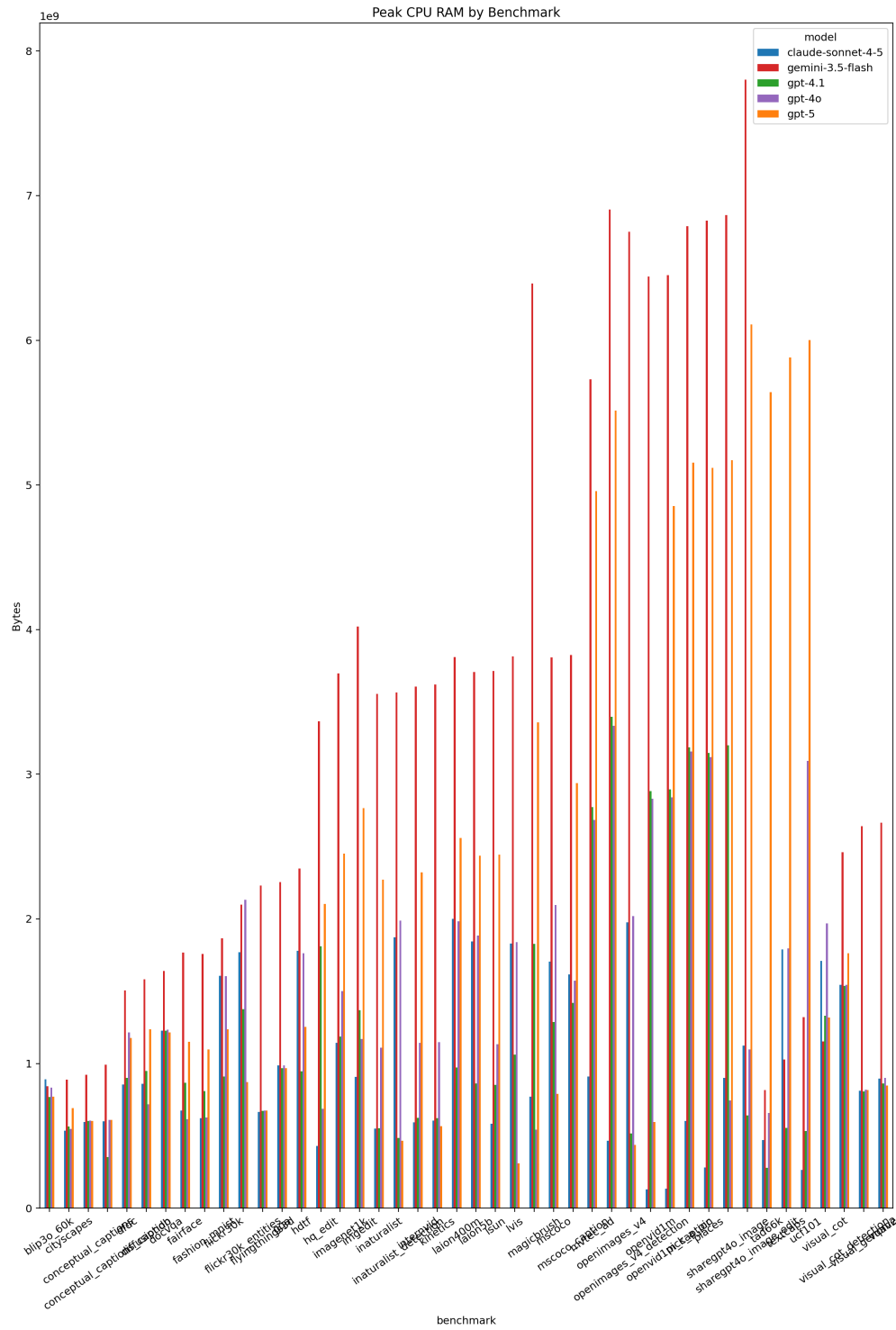


Figure 18: Peak CPU RAM by benchmark and model.

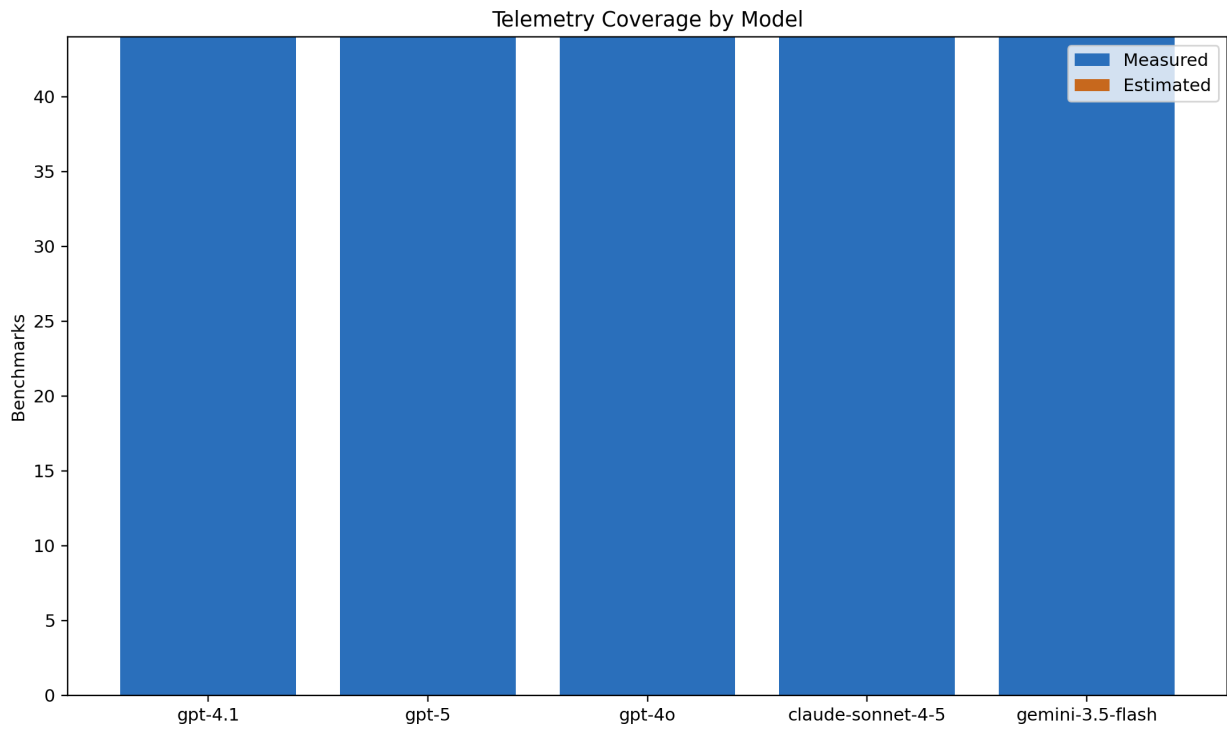


Figure 19: Telemetry coverage by model.

8 Generated Artifacts

The complete generated analysis is saved in `comparison/output/five_api_models_final_clean`. It includes the plots shown above plus the following CSV tables: `benchmark_scores.csv`, `model_summary.csv`, `family_summary.csv`, `benchmark_rankings.csv`, `pairwise_wins.csv`, `bootstrap_summary.csv`, `telemetry_model_summary.csv`, and `telemetry_by_benchmark.csv`.